

# GELU#

<https://pytorch.org/docs/stable/generated/torch.nn.GELU.html>

## Description:

Applies the Gaussian Error Linear Units function. where  $\Phi(x)$  is the Cumulative Distribution Function for Gaussian Distribution. When the approximate argument is 'tanh', Gelu is estimated with: approximate (str, optional) – the gelu approximation algorithm to use: 'none' | 'tanh'. Default: 'none' Input:  $(*)^{(*)}(*)$ , where  $*^{**}$  means any number of dimensions.

## Function Signature

```
class torch.nn.GELU(approximate='none')[source]#
```

Parameters

Shape:

```
extra_repr()[source]#
```

Return type

### Returns:

Tensor

## Code Examples

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```
>>> m = nn.GELU()  
>>> input = torch.randn(2)  
>>> output = m(input)
```

# Detailed Documentation

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## Documentation

Applies the Gaussian Error Linear Units function.

where  $\Phi(x)$  is the Cumulative Distribution Function for Gaussian Distribution.

When the approximate argument is 'tanh', Gelu is estimated with:

approximate (str, optional) – the gelu approximation algorithm to use: 'none' | 'tanh'.

Default: 'none'

Input:  $(*)^{(*)}^{(*)}$ , where  $***$  means any number of dimensions.

Output:  $(*)^{(*)}^{(*)}$ , same shape as the input.

Return the extra representation of the module.

Runs the forward pass.